Preliminary Analysis

Introduction and Data:

Access to educational opportunities can vary significantly between schools, raising questions about how student demographics and school characteristics influence academic participation. In particular, participation in accelerated courses, like Advanced Placement (AP) and International Baccalaureate (IB), and standardized exams (like the SAT and ACT) may be influenced by factors beyond individual academic ability. In “Understanding Unfulfilled AP Potential Across the Participation Pipeline by Race and Income”, Rodriguez and Hernandez-Hamed examine how racial and income differences affect students' access to AP courses and exams. Their research highlights that inequalities can occur at multiple stages, including initial course availability, enrollment, and exam participation. Similarly, in “Advanced Placement Course Enrollment and School-Level Characteristics”, Barnard-Brak, McGaha-Garnett, and Burley investigate how school characteristics relate to AP course enrollment, emphasizing the role that school environments and demographics can play in access to advanced coursework. In “Putting the 'A' in AP: The Effect of Advanced Placement State Policies on Student Participation and Performance”, Callen and Stoddard further examine how state policies influence AP participation, suggesting that differences in educational regulation may contribute to varying levels of access for students across states.

Building on this research, our project uses school specific government sourced data to investigate three questions: How can standardized exam participation from a school be predicted by its characteristics and demographic information? Is participation in accelerated courses, such as AP and IB, associated with student demographic backgrounds? Finally, how does the relative importance of these demographic and school characteristics vary across states? We expect factors

such as school enrollment, school type, and student racial and gender composition to help explain differences in participation. By comparing data and results from Massachusetts, New Mexico, California, and Texas, we aim to identify patterns in educational access and better understand how school characteristics and demographics relate to educational opportunities across different states.

As schools increasingly use data and algorithms to inform decisions about education, it is important to understand how student demographics and school characteristics contribute to educational inequalities. We decided to evaluate large-scale demographic data about school characteristics in order to evaluate how school-level statistics for standardized examination and accelerated coursework may be affected by demographic characteristics. Through gaining an understanding of the data and its biases, we can then critically analyze biases inherent to algorithmic decisions about education through understanding what exactly may feed into an algorithm.

The data we used for research is from the Civil Rights Data Collection (CRDC) by the Office for Civil Rights from the U.S. Department of Education. Collected in 2023 and 2024, it was then released in August of 2026 to the public. The data itself is collected through a survey, required by the U.S. Department of Education's Office for Civil Rights for all schools that receive funding from the federal department. The results in the form of mainly numerical data, and for our purposes pertaining to demographics and characteristics, are open with unrestricted access, and can be used to investigate discrimination complaints, determine if federal civil rights laws have been violated, and generally to provide policy guidance and technical assistance to school districts as well as any other users interested in the information. From the collection, we used files specific to Advanced Placement and International Baccalaureate classes, as well as

SAT/ACT examinations. These detail the number of students from each school who participated in them, broken up by their race, ethnicity, and gender demographics for example. We also utilized files of school characteristics, which encompasses size and learning type for example, and enrollment data per school. From these, we selected 4 states to focus on, with the intent of gaining varying data reflecting the different ends of the education quality spectrum they are on, as well as a population size, to broaden the scope of our research. US News consistently ranks Massachusetts highly in their K-12 overall best education list, currently in second place out of 50, and New Mexico as the 50th place, therefore leading us to select both of these states. The list measures state performance across the time students spend in the school systems, including preschool enrollment, standardized test scores among eighth-graders, high school graduation rate and college readiness. We additionally wanted to look at states with high population and diversity to glean the most amount of information we could regarding students from different backgrounds. The US Census Bureau records the 2025 populations of California and Texas as 39,355,309 and 31,709,821 respectively, making them the two most populous states in the country, and a useful focus for our research. The full data was separated into dataframes relevant to each state. In addition, with data from the 2010 and 2020 censuses, the Bureau has a Diversity Index, which measures the likelihood that two people chosen at random will be from different race and ethnicity groups. Calculated by state, they gave California a 69.7% index, and Texas a 67.0%, making them both in the top 8 states with the highest percentages. We further cleaned these narrowed dataframes to rename column titles to line up with the provided key file by the CRDC to make reading more accessible. For initial processing, all variables with 100% missing values were removed, which included the variables counting the number of nonbinary students. Indicator variables for school type—such as magnet, fully remote, special education—were

changed to a binary numerical format in order for them to be included in models. Additionally, variables from the enrollment dataset were compared to the SAT/ACT, AP, and IB datasets in order to calculate proportions—total, by sex, and by race—of students participating in standardized exams, AP courses, and IB courses.

Methods:

In order to look at placement and participation in accelerated courses relating to the backgrounds of students, we first looked at proportions by schools for SAT/ACT, AP, and IB test files separately and for each selected state. Descriptive statistics for numerical variables were created by state for each of the three datasets, encompassing all of the demographics provided. This lets us measure each state on a numerical level up against each other, to create a context for us to view our results. Variables were created for the proportion of non-white population to total population by school in each state per test file. Similar proportion variables were made for the total number of English learner students to the total population of students taking the tests by school. A variable was also made for the proportion of IDEA (Individuals with Disabilities Education Act) students in the examination files, diagnosed within one of 13 specific disability categories (such as a specific learning disability, autism, ADHD, speech and language impairments, etc.). All of these proportions were then visualized with histograms to compare patterns throughout states, to measure relevance and importance of demographics as far as placement varies between them.

As part of our analysis, we are beginning with running predictive models on the SAT/ACT dataset, specifically the subset with schools in California. Once we find models that adequately describe the data, we will then apply them to the other states and the AP and IB

datasets. We are running Linear Regression and Regression Decision Trees—a machine learning algorithm splits data by conditions of features such as value of a numerical variable and class of a categorical variable—as these models not only give us a metric for how accurately our outcome can be predicted, but also allow us to gain insight on the relative importance of different features of the data to our specified outcome.

Results:

We first ran Linear Regression, with the outcome being the number of students taking the SAT or ACT in a given school and the features being the school's demographic characteristics, such as total number of students identifying with different racial groups and with different disability statuses, as well as the binary indicators for school type. The statistically significant features for this model were ungraded school, charter school, total number of Black male students, total number of IDEA male students, and total number of 504 female students, with negative, negative, positive, negative, and positive coefficients respectively. However, this particular model is not the best predictor of the number of students participating in the SAT or ACT, as the r-squared value was only 0.314, meaning that approximately only 31.4% of variation can be explained by the model.

We then ran a Decision Tree using the same features and outcome. The model had a root mean square error of around 147.2 students. The first node split was the number of female students, and the next variable split for both branches of this initial split was the number of Hispanic male students. While this may give insight into which specific demographic characteristics might be correlated with participation in the SAT or ACT, it is also important to

consider that this model specifically looks at a quantity, and thus larger schools would be predicted to have more participating students by default.

Discussion:

Based on the models we ran, demographic factors do appear to be related to the number of students participating in the SAT or ACT, but the performance of our models indicates that further exploration is needed. As we were using models to predict a quantity, the analysis showed us that larger numbers of students of certain demographic identities were correlated with higher participation in examinations, with the exception of the number of IDEA male students in the Linear Regression model. However, raw quantities are not the most indicative, as the size of schools varies quite a bit across the data. As such, we plan to restructure our analysis to instead evaluate the proportion of students participating in standardized examinations and accelerated courses, and see how the differences in the specific demographic breakdown of each school may impact its such proportions.

Our findings connect to the research of Rodriguez and Hernandez-Hamed, who show that racial and income differences influence AP participation at multiple stages, including course availability, enrollment, and exam participation. Similarly, our results suggest that demographic characteristics are associated with standardized exam participation, although they do not fully explain differences between schools. Barnard-Brak, McGaha-Garnett, and Burley emphasize the importance of school-level characteristics when examining AP enrollment. This is reflected in our Linear Regression results, where school type indicators, including charter and ungraded schools, were significant predictors of SAT/ACT participation. Their research supports the idea that educational opportunities are influenced by both student demographics and the

characteristics of the schools they attend. Finally, Callen and Stoddard demonstrate that state policies, such as requiring AP course offerings and providing exam fee waivers, can influence student participation. This supports our decision to compare multiple states, as differences in educational policies may help explain variations in participation and the relative importance of different school characteristics. Overall, our preliminary results are consistent with previous research suggesting that both demographics and school environments are associated with educational participation, and we will further our models to continue this research.

References

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